VIGNESHKUMAR S

Coimbatore, India
vickylm258@gmail.com — +91 6380045841
Source Code — Portfolio — LinkedIn

Professional Summary

Backend and ML systems developer experienced in building scalable Python APIs, Kubernetes-based infrastructure, and machine learning deployment pipelines. Built predictive autoscaling systems, data ingestion pipelines, and local-first backend services using FastAPI, Docker, PostgreSQL, and PyTorch. Interested in infrastructure engineering, MLOps, and reliable distributed system design.

Technical Skills

- **Languages:** Python, C
- **Backend Development:** FastAPI, REST APIs, Async Processing
- **Machine Learning:** PyTorch, Scikit-Learn, Time-Series Forecasting
- **Databases:** PostgreSQL, MySQL, SQLite
- **Infrastructure & DevOps:** Docker, Kubernetes, Git, Linux
- **Systems & Tools:** Bash, Shell Scripting, CLI Tooling

Projects

AutoScale — Predictive Kubernetes Autoscaling System *Source Code*

- Built a predictive Kubernetes autoscaling prototype using a PyTorch GRU forecasting model capable of forecasting workload trends up to 12 simulation ticks ahead.
- Implemented error-bound tracking, confidence interval estimation, and traffic burst detection to improve scaling stability during sudden workload spikes exceeding 300 RPS.
- Developed a Python-based scaling service integrated with Kubernetes deployments and a fast-forward simulation engine for evaluating autoscaling policies under dynamically changing traffic patterns.

Wiki-ML-Autoscale — Kubernetes-Native MLOps Forecasting System Source Code

- Built a Kubernetes-native MLOps system that fetches Wikipedia pageview data, trains autoregressive forecasting models, and serves predictions through FastAPI REST APIs.
- Deployed automated retraining and inference workflows using Kubernetes CronJobs, Horizontal Pod Autoscaler (HPA), and PersistentVolumes for model persistence and scalable API serving.
- Implemented model evaluation and promotion logic using MAE-based validation, automatically replacing production models only when newly trained versions achieved better forecasting performance.

TrueTrack — Local-First Music Ingestion System GitHub

- Built a self-hosted local-first ingestion system using a FastAPI backend, asynchronous worker pipeline, and Next.js web interface for managing music processing workflows.
- Implemented persistent job state management with SQLite to support resumable workflows that survive restarts, crashes, and partial pipeline failures.
- Developed cross-platform installation and service management tooling for Linux, macOS, and Windows, including CLI-based lifecycle management for local ingestion services.

Education

B.Tech in Artificial Intelligence and Machine Learning Expected April 2027
Sri Shakthi Institute of Engineering and Technology CGPA: 7.95/10